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PINK: physical-informed machine learning for lattice thermal conductivity

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Abstract

Lattice thermal conductivity (κ_L) is crucial for efficient thermal management in electronics and energy conversion technologies. Traditional methods for predicting κ_L are often computationally expensive, limiting their scalability for large-scale material screening. Empirical models, such as the Slack model, offer faster alternatives but require time-consuming calculations for key parameters such as sound velocity and the Grüneisen parameter. This work presents a high-throughput framework, physical-informed kappa (PINK), which combines the predictive power of crystal graph convolutional neural networks (CGCNNs) with the physical interpretability of the Slack model to predict κ_L directly from crystallographic information files (CIFs). Unlike previous approaches, PINK enables rapid, batch predictions by extracting material properties such as bulk and shear modulus from CIFs using a well-trained CGCNN model. These properties are then used to compute the necessary parameters for κ_L calculation through a simplified physical formula. PINK was applied to a dataset of 377,221 stable materials, enabling the efficient identification of promising candidates with ultralow κ_L values, such as $\text{Ag}_3\text{Te}_4\text{W}$ and $\text{Ag}_3\text{Te}_4\text{Ta}$. The platform, accessible via a user-friendly interface, offers an unprecedented combination of speed, accuracy, and scalability, significantly accelerating material discovery for thermal management and energy conversion applications.

Keywords: Physical-informed machine learning, thermoelectrics, lattice thermal conductivity, phonon engineering



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